

Self similar diffusion

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1 The self-similar framework

This section collects the machinery that is common to every equation treated in this document. It is stated once, for a d -dimensional vector field, and each equation section (§4, §5) supplies only the three things that distinguish it: its clock τ , its physical right-hand side R_i^* , and its constraint functionals.

1.1 Rescaling into a self-similar frame

Let $u_i^*(\mathbf{x}^*, t^*)$, $i = 1, \dots, d$, satisfy an evolution equation of the form

$$\frac{\partial u_i^*}{\partial t^*} = R_i^*(u^*), \quad (1)$$

on $\mathbb{R}^d$. We rescale amplitude, length, and time by

$$u_i = \frac{u_i^*}{U}, \quad x_j = \frac{x_j^*}{L}, \quad t = \frac{t^*}{s(t)}, \quad (2)$$

where the amplitude scale U and the length scale L depend on the non dimensional time t . The timescale is determined by specifying the clock

$$\tau = \frac{dt^*}{dt}, \quad (3)$$

The clock τ is problem specific and can contain the time dependent length and amplitude scales. It is defined in §4.1 and §5.1 for the two cases considered here.

Throughout, a prime denotes differentiation with respect to t and we use the following abbreviations

$$\sigma_L = \frac{L'}{L}, \quad \sigma_U = \frac{U'}{U}. \quad (4)$$

Differentiating $x_j = x_j^*/L$ with respect to t at fixed x_j^* gives, for each component j ,

$$\frac{dx_j}{dt} = -x_j^* \frac{L'}{L^2} = -x_j \sigma_L. \quad (5)$$

Applying the chain rule to the left-hand side of (1), and using (3) and (5),

$$\begin{aligned} \frac{\partial u_i^*}{\partial t^*} &= \frac{dt}{dt^*} \left(\frac{\partial(Uu_i)}{\partial t} + U \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} \right) \\ &= \frac{U}{\tau} \left(\frac{\partial u_i}{\partial t} + \sigma_U u_i - \sigma_L x_j \frac{\partial u_i}{\partial x_j} \right). \end{aligned} \quad (6)$$

Equating (6) to R_i^* and solving for $\partial u_i/\partial t$ gives the result used throughout this document. Under the rescaling (2), the equation (1) becomes

$$\frac{\partial u_i}{\partial t} = \underbrace{\sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i}_{\text{scaling terms}} + \underbrace{\frac{\tau}{U} R_i^*(Uu, Lx)}_{\text{physical terms}}. \quad (7)$$

Accordingly we split (7) into two flows. The *scaling operator* $\Phi_{\Delta t}^{(S)}$ advances the scaling terms,

$$\Phi_{\Delta t}^{(S)}(u_i) = u_i + \int_t^{t+\Delta t} \left(\sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i \right) dt, \quad (8)$$

and the *physics operator* $\Phi_{\Delta t}^{(P)}$ advances the rest,

$$\Phi_{\Delta t}^{(P)}(u_i) = u_i + \int_t^{t+\Delta t} \frac{\tau}{U} R_i^*(Uu, Lx) dt. \quad (9)$$

Both are written as the field plus the integral of the terms they advance. $\Phi^{(S)}$ is obtained in closed form below and is shared; $\Phi^{(P)}$ is equation-specific and is supplied in §4.2 and §5.5.

1.2 The scaling operator

Evaluating $\Phi_{\Delta t}^{(S)}$ amounts to solving

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i. \quad (10)$$

This is a linear, first-order PDE and can be solved by the method of characteristics. Note that the scaling terms do not couple the components of u_i , so the argument below is componentwise.

Let $x_j(t)$ denote a characteristic curve satisfying

$$\frac{dx_j}{dt} = -x_j \sigma_L, \quad (11)$$

which is (5): the characteristics are exactly the paths of fixed physical position x_j^* . The total derivative of u_i along such a curve is

$$\frac{du_i}{dt} = \frac{\partial u_i}{\partial t} + \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} = \frac{\partial u_i}{\partial t} - \sigma_L x_j \frac{\partial u_i}{\partial x_j}, \quad (12)$$

so along the characteristic (10) reduces to the ordinary differential equation

$$\frac{du_i}{dt} = -\sigma_U u_i. \quad (13)$$

Integrating (11) from t_k to t_{k+1} gives

$$\ln x_j(t_{k+1}) - \ln x_j(t_k) = -[\ln L_{k+1} - \ln L_k] \implies x_j(t_{k+1}) = x_j(t_k) \frac{L_k}{L_{k+1}}, \quad (14)$$

so that the characteristic arriving at the evaluation point $x_j \equiv x_j(t_{k+1})$ originated at

$$x_j(t_k) = x_j \frac{L_{k+1}}{L_k}. \quad (15)$$

Integrating (13) in the same way gives $u_i(t_{k+1}) = u_i(t_k) U_k/U_{k+1}$. Combining this with the characteristic origin (15) and the initial condition $u_i(t_k, \cdot)$ gives the scaling operator in closed form,

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{U_k}{U_{k+1}} u_i \left(\mathbf{x} \frac{L_{k+1}}{L_k} \right). \quad (16)$$

Writing the step ratios

$$\tilde{U}_{k+1} = \frac{U_{k+1}}{U_k}, \quad \tilde{L}_{k+1} = \frac{L_{k+1}}{L_k}, \quad (17)$$

this is

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{1}{\tilde{U}_{k+1}} u_i \left(\mathbf{x} \tilde{L}_{k+1} \right). \quad (18)$$

The operator depends on Δt only through the ratios $\tilde{U}_{k+1}, \tilde{L}_{k+1}$; given those, no time integration remains.

The scaling terms in $\mathbf{k}$ -space. For equations formulated spectrally it is useful to have the transform of the scaling terms. Using $\widehat{\cdot}$ for the spatial Fourier transform,

$$\hat{u}_i(\mathbf{k}) = \int u_i(\mathbf{x}) e^{-i\mathbf{k}_j x_j} d\mathbf{x}, \quad (19)$$

and writing the spread term in conservative form,

$$x_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} (x_j u_i) - d u_i, \quad (20)$$

gives

$$\widehat{x_j \frac{\partial u_i}{\partial x_j}} = i k_j \left(i \frac{\partial \hat{u}_i}{\partial k_j} \right) - d \hat{u}_i = - \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + d \hat{u}_i \right), \quad (21)$$

so that the transform of (10) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + d \hat{u}_i \right) - \sigma_U \hat{u}_i. \quad (22)$$

Rather than solving (22) directly in $\mathbf{k}$ -space, it is easier to transform to physical space, apply (18), and transform back; the Fourier transform (19) is linear, so the two are equivalent.

1.3 Spectral interpolation

The closed form (18) requires u_i at the rescaled points $\mathbf{x} \tilde{L}_{k+1}$, which are not grid points. On a uniform periodic grid the represented field is band-limited, so it can be evaluated at arbitrary points exactly, with no interpolation error, by summing its truncated Fourier series there:

$$u_i(\mathbf{x} \tilde{L}_{k+1}) = \sum_{\mathbf{k} \in \mathcal{K}} \hat{u}_i(\mathbf{k}) \exp(i \mathbf{k}_j x_j \tilde{L}_{k+1}), \quad (23)$$

where $\mathcal{K}$ is the set of represented wavenumbers and $\hat{u}_i(\mathbf{k})$ are the normalized transform coefficients of u_i . This is the defining property of the scheme: the scaling step introduces no spatial discretization error of its own.

1.4 Constraint functionals

Nothing so far fixes the ratios $\tilde{U}_{k+1}, \tilde{L}_{k+1}$ that the scaling step (18) depends on. They are fixed by requiring the rescaled field to carry prescribed values of two functionals F_1, F_2 of the field, each homogeneous under (18): there exist exponents p_m, q_m with

$$F_m \left[\frac{1}{\tilde{U}} u(\cdot \tilde{L}) \right] = \tilde{U}^{-p_m} \tilde{L}^{-q_m} F_m[u], \quad m = 1, 2. \quad (24)$$

The targets F_1, F_2 are fixed once, from the initial condition (§2.2). Given a field u with current values $F_m^* = F_m[u]$, imposing $\tilde{U}^{-p_m} \tilde{L}^{-q_m} F_m^* = F_m$ and taking logarithms gives a system linear in the log-ratios,

$$\begin{bmatrix} p_1 & q_1 \\ p_2 & q_2 \end{bmatrix} \begin{bmatrix} \ln \tilde{U} \\ \ln \tilde{L} \end{bmatrix} = \begin{bmatrix} \ln F_1^* - \ln F_1 \\ \ln F_2^* - \ln F_2 \end{bmatrix}, \quad (25)$$

uniquely solvable precisely when

$$p_1 q_2 - p_2 q_1 \neq 0, \quad (26)$$

that is, when the two functionals respond independently to amplitude and to length rescaling. The functionals themselves are equation-specific (§4.3, §5.6); the substep that applies (25) is CONSTRAINEDSCALE (§2.4).

1.5 Power-law ansatz and the step in physical time

The algorithm also needs to *predict* scale ratios over a fraction of a step, at points where no constraint is available to fix them (§2.6). For this we assume that both scales follow a power law in physical time,

$$X = (t^* + \alpha_X)^{\beta_X}, \quad X \in \{U, L\}, \quad (27)$$

so that the ratio over a fraction i of step k is

$$\frac{X_{k+i}}{X_k} = \left(1 + \frac{i \Delta t_k^*}{t_k^* + \alpha_X}\right)^{\beta_X}. \quad (28)$$

Since the time step Δt is taken in the self-similar time t , the corresponding physical increment must be obtained from the clock,

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau \, dt \approx \frac{\Delta t}{12} (23\tau_k - 16\tau_{k-1} + 5\tau_{k-2}), \quad (29)$$

using a three-point Adams–Bashforth-style quadrature on the stored history of τ . Eliminating $t_k^* + \alpha_X = X_k^{1/\beta_X}$ from (27) and taking logarithms gives the function that the algorithm uses in both directions,

$$\ln \frac{X_{k+i}}{X_k} = G_X(i, \beta_X; k), \quad G_X(i, \beta; k) = \beta \ln \left(1 + \exp \left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k}{\beta} \right)\right). \quad (30)$$

It is evaluated forward, with $i = \frac{1}{2}$, to predict a half-step ratio (§2.4), and inverted for β , with $i = 1$, to estimate the exponents from the realized history (§2.7).

Logarithmic arithmetic. U_k and L_k may become extremely large or small, so we store $\ln U_k$ and $\ln L_k$ rather than U_k and L_k , and perform all arithmetic on them in the logarithmic number system, where multiplication and division become addition and subtraction, and where sums and differences are evaluated by

$$\log_b(X + Y) = \log_b(X) + \log_b(1 + b^{y-x}), \quad \log_b(X - Y) = \log_b(X) + \log_b(1 - b^{y-x}) \quad (X > Y > 0), \quad (31)$$

with $x = \log_b(X)$, $y = \log_b(Y)$. In particular (29) is evaluated as

$$\ln \Delta t_k^* = \ln \frac{\Delta t}{12} + \ln(23 e^{\ln \tau_k} - 16 e^{\ln \tau_{k-1}} + 5 e^{\ln \tau_{k-2}}), \quad (32)$$

with the sum taken via (31), and each $\ln \tau_k$ supplied in log form by the equation at hand ((56), (75)). Exponentiation is deferred until a value is actually needed, e.g. for output.

2 Algorithm

The integrator below is stated once, for any equation that supplies the three items of §1: a clock τ , a physics operator $\Phi^{(P)}$, and a pair of constraint functionals. Everything else — the splitting, the scale bookkeeping, the exponent estimation — is common.

2.1 Domain, grid, and notation

The spatial domain is periodic with a fixed physical side length H and N grid points per direction, centred at the origin, so that $x_j \in [-H/2, H/2)$ with uniform spacing $\Delta x = H/N$ and grid points

$$x_n = -\frac{H}{2} + n \Delta x, \quad n = 0, 1, \dots, N-1 \quad (33)$$

in each of the d directions. Every integral over the domain below is approximated by the Riemann sum

$$\int f(\mathbf{x}) \, d\mathbf{x} \approx (\Delta x)^d \sum_{\mathbf{n}} f(\mathbf{x}_{\mathbf{n}}) \quad (34)$$

on this grid.

An index range abbreviates the corresponding tuple of stored values, e.g. $\ln \tau_{k-2:k} = (\ln \tau_{k-2}, \ln \tau_{k-1}, \ln \tau_k)$. The run constants $\Delta t, H, N$, the constraint targets F_1, F_2 , and whatever physical parameters the equation carries are visible to every subroutine and are omitted from their signatures. As set out in §1.5, all arithmetic on the scales is done on $\ln U_k, \ln L_k$.

2.2 Initialization

A run is specified by the initial field u_0 , the domain parameters H, N , a fixed self-similar timestep Δt , a time horizon T_{end} , the splitting scheme $scheme \in \{\text{LIETROTTER}, \text{STRANG}\}$ (§2.5, §2.6), and initial guesses $\beta_U^{(0)}, \beta_L^{(0)}$, which only seed the bracket of the first root-find for β_U, β_L (§2.7). From these:

- The number of steps is $N_{\text{steps}} = \lceil T_{\text{end}}/\Delta t \rceil$.
- The self-similar frame is initialized at the initial condition itself, $L_0 = U_0 = 1$, i.e. $\ln L_0 = \ln U_0 = 0$.
- The *target* values F_1, F_2 of the constraint functionals (§1.4) are fixed once, from u_0 itself,

$$F_1 = F_1[u_0], \quad F_2 = F_2[u_0]. \quad (35)$$

- The flag β_{ok} , which records whether the current β_U, β_L are usable by STRANG, starts as $\beta_{\text{ok}} = \text{false}$.

2.3 Top-level integrator

Algorithm 1 Top-level integrator

Require: $u_0, H, N, \Delta t, T_{\text{end}}, scheme, \beta_U^{(0)}, \beta_L^{(0)}$

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1:  $N_{\text{steps}} \leftarrow \lceil T_{\text{end}}/\Delta t \rceil$ 
2:  $\ln U_0 \leftarrow 0, \ln L_0 \leftarrow 0$ 
3: Compute constraint targets  $F_1, F_2$  from  $u_0$  via (35)
4:  $\beta_U \leftarrow \beta_U^{(0)}, \beta_L \leftarrow \beta_L^{(0)}, \beta_{\text{ok}} \leftarrow \text{false}$ 
5: Store  $(u_0, \ln U_0, \ln L_0, \ln \tau_0, \beta_U, \beta_L, \beta_{\text{ok}})$ 
6: for  $k = 0, 1, \dots, N_{\text{steps}} - 1$  do
7:   if  $scheme = \text{STRANG}$  and  $\beta_{\text{ok}}$  then
8:      $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{STRANG}(u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \beta_U, \beta_L)$  ▷ §2.6
9:   else
10:     $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{LIETROTTER}(u_k, \ln U_k, \ln L_k)$  ▷ §2.5
11:   end if
12:    $u_{k+1} \leftarrow \text{APPLYBC}(u_{k+1})$  ▷ equation-specific, §2.4
13:    $\ln \tau_{k+1} \leftarrow \ln \tau(\ln U_{k+1}, \ln L_{k+1})$  ▷ (56), (75)
14:   if  $k + 1 \geq 3$  then
15:      $(\beta_U, \beta_L, \beta_{\text{ok}}) \leftarrow \text{ESTIMATEBETA}(\ln U_{k:k+1}, \ln L_{k:k+1}, \ln \tau_{k-2:k}, \beta_U, \beta_L)$  ▷ §2.7
16:   end if
17:   Store  $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}, \ln \tau_{k+1}, \beta_U, \beta_L, \beta_{\text{ok}})$ 
18: end for
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The run parameter *scheme* selects the step function. STRANG needs usable estimates of β_U, β_L , so the loop falls back to LIETROTTER whenever β_{ok} is false. This is always the case for the first three steps $k = 0, 1, 2$: the first β update needs four stored states, so it first runs at the end of step $k = 2$. It is also the case for every step that follows a failed β update (§2.7), until a later update succeeds. The β update itself runs regardless of *scheme*; under *scheme* = LIETROTTER the estimates are diagnostic only.

2.4 Substeps

Both splitting schemes are assembled from the substeps below. None of them applies the boundary treatment, which is handled only by the top-level loop.

Evolve($u, \Delta t$). Applies the physics operator $\Phi_{\Delta t}^{(P)}$ of (9): it advances the non-scaling terms of (7) over Δt , leaving the scales $\ln U, \ln L$ untouched. This is the one substep the framework does not supply. Its realization is equation-specific (§4.2, §5.5), and it is required to be at least first-order accurate for the Lie–Trotter composition and at least second-order accurate for the Strang composition, since the splitting cannot recover an order the substep does not have.

Scale($u, \ln \tilde{U}, \ln \tilde{L}$). Applies the scaling operator (18) with prescribed ratios $\tilde{U}, \tilde{L}$, returning $\tilde{U}^{-1}u(\mathbf{x}_n \tilde{L})$ at each grid point, evaluated with the spectral interpolation of §1.3. Here, and wherever the substeps are used, $\tilde{U}$ and $\tilde{L}$ denote the ratio of U , respectively L , after the substep to its value before it. Inside a Lie–Trotter step this is $\tilde{U}_{k+1} = U_{k+1}/U_k$ of (17); inside a Strang step it is a ratio across a half step, $U_{k+1/2}/U_k$ or $U_{k+1}/U_{k+1/2}$ (§2.6).

ConstrainedScale($u, \ln U, \ln L$). Rescales u so that the result carries the fixed targets F_1, F_2 of (35), and advances the log-scale state accordingly:

1. **Functionals of the input**, $F_1^* = F_1[u]$, $F_2^* = F_2[u]$, evaluated on the grid via (34).
2. **Ratios** $\ln \tilde{U}, \ln \tilde{L}$ from the constraint system (25).
3. **Rescale**, $u' = \text{SCALE}(u, \ln \tilde{U}, \ln \tilde{L})$.
4. **Return** u' together with the advanced state $\ln U + \ln \tilde{U}, \ln L + \ln \tilde{L}$.

No time increment enters **CONSTRAINEDSCALE**: the ratios are fixed by the functionals of the field it is given, not by the elapsed time. The same substep therefore serves as the full scaling step of Lie–Trotter and as the closing scaling half step of Strang.

PredictScale($i, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \beta_U, \beta_L$). Predicts the ratios over the fraction i of step k from the power-law form (27) rather than from the constraints. It evaluates $\ln \Delta t_k^*$ from (32) and returns $(G_U(i, \beta_U; k), G_L(i, \beta_L; k))$ using (30). The same function G_X is inverted for β in §2.7.

ApplyBC(u). Imposes whatever boundary treatment the equation requires, after the full step has been assembled. It is the identity for a genuinely periodic problem; the 1D diffusion equation instead clamps the domain edges (§4.5).

2.5 LieTrotter: Lie–Trotter splitting

A Lie–Trotter step composes a full physics step with a full scaling step, $u_{k+1} = \Phi_{\Delta t}^{(S)}(\Phi_{\Delta t}^{(P)}(u_k))$, with the scaling ratios fixed by the constraints. It uses no estimate of β_U, β_L , so it is valid from the first step. It is written for a general input state $(u, \ln U, \ln L)$ rather than only a stored state $(u_k, \ln U_k, \ln L_k)$, so that **STRANG** can call it on a half-stepped state.

Algorithm 2 Lie–Trotter step

- 1: **function** LIETROTTER($u, \ln U, \ln L$)
 - 2: $u^{(P)} \leftarrow \text{EVOLVE}(u, \Delta t)$ ▷ §2.4
 - 3: **return** CONSTRAINEDSCALE($u^{(P)}, \ln U, \ln L$) ▷ §2.4
 - 4: **end function**
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2.6 Strang: Strang splitting

A Strang step is the symmetric composition

$$u_{k+1} = \Phi_{\Delta t/2}^{(S)} \left(\Phi_{\Delta t}^{(P)} \left(\Phi_{\Delta t/2}^{(S)}(u_k) \right) \right) \quad (36)$$

of a scaling half step, a full physics step, and a second scaling half step. The last two of these are exactly a Lie–Trotter step — a full physics step followed by a scaling step whose ratios come from the constraints — so STRANG is a predicted scaling half step followed by a call to LIETROTTER.

Algorithm 3 Strang step

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1: function STRANG( $u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \beta_U, \beta_L$ )
2:    $(\ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k)) \leftarrow \text{PREDICTSCALE}(\frac{1}{2}, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \beta_U, \beta_L)$  ▷ (37)
3:    $u_{k+1/2} \leftarrow \text{SCALE}(u_k, \ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k))$ 
4:    $\ln U_{k+1/2} \leftarrow \ln U_k + \ln(U_{k+1/2}/U_k); \ln L_{k+1/2} \leftarrow \ln L_k + \ln(L_{k+1/2}/L_k)$ 
5:   return LIETROTTER( $u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2}$ ) ▷ §2.5
6: end function

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No constraint fixes the state at $t_k + \Delta t/2$, so the first scaling half step takes its ratios from the current power-law estimates, via PREDICTSCALE with $i = 1/2$:

$$\ln \frac{U_{k+1/2}}{U_k} = G_U(\tfrac{1}{2}, \beta_U; k), \quad \ln \frac{L_{k+1/2}}{L_k} = G_L(\tfrac{1}{2}, \beta_L; k). \quad (37)$$

The call to LIETROTTER then advances $u_{k+1/2}$ over the full Δt , and its CONSTRAINEDSCALE substep — the second scaling half step — sets $U_{k+1}/U_{k+1/2}$ and $L_{k+1}/L_{k+1/2}$ so that u_{k+1} satisfies the constraints. The half-step state $(u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2})$ is internal to the step and is not stored. The top-level loop calls STRANG only when β_{ok} is true (§2.3), which guarantees that β_U, β_L were fitted to the stored history at the end of the previous step and that $\ln \tau_{k-2}$ exists.

2.7 EstimateBeta: estimating β_U and β_L

After every step with $k+1 \geq 3$, regardless of *scheme*, β_U and β_L are re-estimated from the log-scale history. Four stored states are needed: $\ln X_k$ and $\ln X_{k+1}$ for the ratio, and $\ln \tau_{k-2:k}$ for Δt_k^* . The update inverts the function G_X of (30) with $i = 1$, the same function that PREDICTSCALE evaluates forward with $i = 1/2$:

$$G_U(1, \beta_U; k) = \ln U_{k+1} - \ln U_k, \quad (38)$$

and identically with $U \rightarrow L$ for β_L . Each of (38) and its L -counterpart is a single scalar equation in a single unknown (β_U , resp. β_L), solved by **brentq**; the implementer will need to choose a bracketing interval (e.g. centred on the previous β_U, β_L estimate — on $\beta_U^{(0)}, \beta_L^{(0)}$ for the first update — widened adaptively if it fails to bracket a root).

If both root-finds succeed, ESTIMATEBETA returns the new β_U, β_L with $\beta_{\text{ok}} = \text{true}$. If either fails to bracket a root even after widening, the failed estimate keeps its previous value (which seeds the next bracket), and ESTIMATEBETA returns $\beta_{\text{ok}} = \text{false}$, so the next step falls back to LIETROTTER. Before $k+1 \geq 3$ no update is made: β_U, β_L stay at $\beta_U^{(0)}, \beta_L^{(0)}$ and β_{ok} stays false.

3 Post-processing

A completed run leaves behind the log-scale history $\ln U_k, \ln L_k$ and the clock history $\ln \tau_k$. From these we recover the power-law parameters α_X, β_X of the ansatz (27). This is the offline, global counterpart of ESTIMATEBETA (§2.7), which fits β_X one step at a time; here the whole history is used at once, so the fit also serves as a check on the run.

Write

$$\mathcal{L}_k = \ln L_k, \quad \mathcal{U}_k = \ln U_k, \quad (39)$$

together with the per-step log-ratios

$$\ell_k = \ln L_{k+1} - \ln L_k, \quad \lambda_k = \ln U_{k+1} - \ln U_k, \quad (40)$$

which for a Lie–Trotter step equal $\ln \tilde{L}_{k+1}$ and $\ln \tilde{U}_{k+1}$ of (17), and for a Strang step combine both half-step ratios. The stored scales are their cumulative sums,

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^k \ell_i, \quad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^k \lambda_i, \quad (41)$$

which grow only linearly in k ; unlike L_k and U_k themselves, $\mathcal{L}_k$ and $\mathcal{U}_k$ never overflow.

3.1 The scaling rates under the ansatz

Differentiating the ansatz (27) with respect to t , and using the clock (3) to convert from t^* , the logarithmic rates (4) are

$$\sigma_X = \frac{1}{X} \frac{dX}{dt^*} \frac{dt^*}{dt} = \frac{\beta_X \tau}{t^* + \alpha_X}, \quad X \in \{U, L\}. \quad (42)$$

Inverting the ansatz to eliminate the unknown offset,

$$t^* + \alpha_X = X^{1/\beta_X}, \quad (43)$$

and substituting into (42) gives the rates in terms of the scales alone,

$$\sigma_X = \beta_X \tau X^{-1/\beta_X}. \quad (44)$$

Taking logarithms,

$$\ln \sigma_{X,k} = \ln \beta_X + \ln \tau_k - \frac{\mathcal{X}_k}{\beta_X}, \quad \mathcal{X}_k = \ln X_k. \quad (45)$$

Finally, the rates themselves are estimated directly from the log-ratios already in hand, without any finite difference of L_k or U_k themselves,

$$\sigma_{L,k} \approx \frac{\ell_k}{\Delta t}, \quad \sigma_{U,k} \approx \frac{\lambda_k}{\Delta t}. \quad (46)$$

3.2 Recovering α_X and β_X

Equation (45) is linear in $\mathcal{X}_k$ once the clock is divided out. Regressing

$$\ln \sigma_{X,k} - \ln \tau_k = \ln \beta_X - \frac{1}{\beta_X} \mathcal{X}_k \quad (47)$$

against $\mathcal{X}_k$ therefore yields β_X twice over: once as $e^{\text{intercept}}$ and once as $-1/\text{slope}$. Agreement between the two is a check that the run is in fact following a power law over the fitted range; disagreement indicates that it is not, or that the range includes the initial transient before the frame has locked on.

With β_X in hand, the offset follows from (43),

$$\alpha_X = X_k^{1/\beta_X} - t_k^*, \quad (48)$$

where the physical time is accumulated from the per-step increments of (29),

$$t_k^* = t_0^* + \sum_{i=0}^{k-1} \Delta t_i^*. \quad (49)$$

In exact arithmetic (48) is independent of k ; in practice it is averaged over the fitted range, and its scatter is a second check on the ansatz.

For the 1D diffusion equation the clock is $\tau = L^2/D$, and (45) specializes to

$$\ln \sigma_{L,k} = \ln \frac{\beta_L}{D} + \left(2 - \frac{1}{\beta_L}\right) \mathcal{L}_k, \quad (50)$$

$$\ln \sigma_{U,k} = \ln \frac{\beta_U}{D} + 2 \mathcal{L}_k - \frac{\mathcal{U}_k}{\beta_U}, \quad (51)$$

so that for $X = L$ the regression (47) collapses onto a single variable, while for $X = U$ it is a two-variable fit in $\mathcal{L}_k$ and $\mathcal{U}_k$.

4 The 1D diffusion equation

The first instantiation of the framework is the one-dimensional diffusion equation, $d = 1$, with a single scalar component; we drop the component index throughout this section. It serves as a testbed: every operator the framework needs is available in closed form, so the only approximation in a run is the operator splitting itself.

4.1 Governing equation and clock

The dimensional equation is

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}, \quad (52)$$

so that, in the notation of (1),

$$R^*(u^*) = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \quad (53)$$

Under the rescaling (2) this term becomes

$$D \frac{\partial^2 (uU)}{\partial (x^*)^2} = DU \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{1}{L} \right) \frac{1}{L} = \frac{DU}{L^2} \frac{\partial^2 u}{\partial x^2}, \quad (54)$$

having used $dx/dx^* = 1/L$. The clock is then chosen to make the physical term of the lemma (7) carry unit coefficient,

$$\frac{\tau}{U} \cdot \frac{DU}{L^2} = 1 \quad \implies \quad \tau = \frac{L^2}{D}, \quad (55)$$

which is the diffusive time scale of the current length scale, as expected. In log form, as required by (32),

$$\ln \tau_k = 2 \ln L_k - \ln D. \quad (56)$$

Substituting (55) into the rescaling lemma (7) gives the equation the algorithm integrates,

$$\frac{\partial u}{\partial t} = \sigma_L x \frac{\partial u}{\partial x} - \sigma_U u + \frac{\partial^2 u}{\partial x^2}. \quad (57)$$

4.2 The physics step

Because the diffusion term in (57) has unit coefficient, the physics operator (9),

$$\Phi_{\Delta t}^{(P)}(u_k) = u_k + \int_{t_k}^{t_k + \Delta t} \frac{\partial^2 u}{\partial x^2} dt, \quad (58)$$

has the closed-form spectral solution

$$\hat{u}_j^{(P)} = \hat{u}_j e^{-k_j^2 \Delta t}, \quad k_j = \frac{2\pi j}{H}, \quad j = 0, \dots, \lfloor N/2 \rfloor, \quad (59)$$

where $\hat{u}_j = \text{RFFT}(u)_j$. This is an exact one-shot spectral propagator, not an approximate time-marching scheme, so no sub-stepping or implicit solve is needed, and the accuracy requirement on EVOLVE (§2.4) is met trivially at any order. Inverse-transforming gives the advanced field $u^{(P)}(x_n)$.

4.3 Moment constraints

The constraint functionals of §1.4 are taken to be the zeroth and second moments of the field,

$$F_1[u] = \int u dx \equiv M_0, \quad F_2[u] = \int x^2 u dx \equiv M_2, \quad (60)$$

so that M_0 measures the amount of the conserved quantity and M_2 its spread. Writing M_0^*, M_2^* for the moments of the field being scaled, and substituting $y = x \tilde{L}$ in each integral, the scaled field carries

$$\int \frac{1}{\tilde{U}} u(x \tilde{L}) dx = \frac{1}{\tilde{U} \tilde{L}} M_0^*, \quad (61)$$

$$\int x^2 \frac{1}{\tilde{U}} u(x \tilde{L}) dx = \frac{1}{\tilde{U} \tilde{L}^3} M_2^*, \quad (62)$$

so the homogeneity exponents of (24) are

$$(p_1, q_1) = (1, 1), \quad (p_2, q_2) = (1, 3), \quad (63)$$

with $p_1 q_2 - p_2 q_1 = 2 \neq 0$, so the constraint system (25) is uniquely solvable: the two moments do respond independently to amplitude and to length rescaling. Solving it,

$$\tilde{L} = \sqrt{\frac{M_0 M_2^*}{M_0^* M_2}}, \quad (64)$$

$$\tilde{U} = \frac{M_0^*}{M_0 \tilde{L}} = \left(\frac{M_0^*}{M_0} \right)^{3/2} \sqrt{\frac{M_2}{M_2^*}}. \quad (65)$$

Open. An earlier form of (64) and (65) omitted the factor M_0 , giving $\tilde{L} = \sqrt{M_2^*/(M_0^* M_2)}$ and $\tilde{U} = (M_0^*)^{3/2} M_0^{-1} \sqrt{M_2/M_2^*}$; the present implementation uses that form. The two agree exactly when $M_0 = 1$, and differ by a factor $M_0^{\pm 1/2}$ otherwise. The discrepancy traces to whether the target second moment is taken raw, as in (60) and (35), or normalized by M_0 . Since diffusion conserves M_0 and the initial condition is in practice normalized to unit mass, the difference has not so far been visible in a run. Which convention is intended should be settled, and the implementation brought into line with it.

4.4 Spectral interpolation on the real transform

For $d = 1$ and a real-valued field, the generic interpolation formula (23) is realized on the non-negative frequency components of a real Fast Fourier Transform (RFFT):

$$\begin{aligned} \Phi_{\Delta t}^{(S)}(u_k)(x) = \frac{1}{\tilde{U}_{k+1}} & \left\{ \hat{u}_0^{(k)} + 2 \sum_{j=1}^{N/2-1} \left[\operatorname{Re}(\hat{u}_j^{(k)}) \cos\left(\frac{2\pi j \tilde{L}_{k+1} x}{H}\right) \right. \right. \\ & \left. \left. - \operatorname{Im}(\hat{u}_j^{(k)}) \sin\left(\frac{2\pi j \tilde{L}_{k+1} x}{H}\right) \right] \right. \\ & \left. + \hat{u}_{N/2}^{(k)} \cos\left(\frac{\pi N \tilde{L}_{k+1} x}{H}\right) \right\}, \end{aligned} \quad (66)$$

where $\hat{u}_j^{(k)} = \frac{1}{N} \text{RFFT}(u_k)_j$ for $j = 0, 1, \dots, N/2$, and the Nyquist term at $j = N/2$ is included only if N is even. The factor of 2 accounts for the Hermitian conjugate modes omitted by the real transform.

4.5 Boundary treatment

The substep APPLYBC of §2.4 imposes a homogeneous Dirichlet condition by clamping the two edge grid points of the centred domain (33) after each completed step,

$$u_{k+1}(x_0) \leftarrow 0, \quad u_{k+1}(x_{N-1}) \leftarrow 0. \quad (67)$$

On the centred grid $x \in [-H/2, H/2]$ these are two distinct grid points, not periodic images of one another.

Open. This sits in tension with the rest of the discretization: the spectral propagator (59) and the spectral interpolation (66) both presume a periodic field, whereas (67) imposes a Dirichlet condition. The

two are consistent only so long as u is negligible near the domain edges, which requires the initial condition to be localized well inside the domain and the run to be stopped before the profile spreads to the boundary. The scaling step works against this — it is what keeps the profile from spreading in the self-similar frame — but the condition is an assumption of the scheme rather than a consequence of it, and is not currently monitored.

5 The Navier–Stokes equations

The second instantiation is the incompressible Navier–Stokes equations, with $d = 3$ and u_i the velocity field. The framework applies unchanged; what this section supplies is the clock, the nondimensional right-hand side, the structure of the physics operator, and the constraint functionals. One item needed to actually run the algorithm of §2 — the time-marching scheme inside $\Phi^{(P)}$ — remains open, and is marked as such below.

5.1 Governing equations and clock

The dimensional equations are

$$\frac{\partial u_i^*}{\partial t^*} + u_j^* \frac{\partial u_i^*}{\partial x_j^*} = -\frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} + \nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*}, \quad (68)$$

together with the continuity equation,

$$\frac{\partial u_i^*}{\partial x_i^*} = 0, \quad (69)$$

so that, in the notation of (1),

$$R_i^*(u^*) = -u_j^* \frac{\partial u_i^*}{\partial x_j^*} - \frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} + \nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*}. \quad (70)$$

In addition to the rescaling (2) of u_i and x_j , the pressure is nondimensionalized by the dynamic pressure,

$$p = \frac{p^*}{\rho U^2}. \quad (71)$$

Term by term, the convective and pressure terms scale as U^2/L ,

$$u_j^* \frac{\partial u_i^*}{\partial x_j^*} = \frac{U^2}{L} u_j \frac{\partial u_i}{\partial x_j}, \quad \frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} = \frac{U^2}{L} \frac{\partial p}{\partial x_i}, \quad (72)$$

while the viscous term scales as $\nu U/L^2$,

$$\nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu U}{L^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (73)$$

The clock is chosen so that the dominant, advective part of the physical term of the lemma (7) carries unit coefficient,

$$\frac{\tau}{U} \cdot \frac{U^2}{L} = 1 \quad \implies \quad \tau = \frac{L}{U}, \quad (74)$$

which is the eddy turnover time of the current scales. In log form, as required by (32),

$$\ln \tau_k = \ln L_k - \ln U_k. \quad (75)$$

Applying $\tau/U = L/U^2$ to (70) and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu}, \quad (76)$$

the rescaling lemma (7) gives the nondimensional momentum equation

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{1}{Re} \frac{\partial^2 u_i}{\partial x_j \partial x_j}, \quad (77)$$

and the continuity equation is unchanged by the rescaling,

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (78)$$

5.2 Viscosity

Let the kinematic viscosity be given by

$$\nu = \nu_{\text{turb}} + \nu_{\text{mol}}, \quad (79)$$

where ν_{mol} is the molecular viscosity and ν_{turb} is a turbulent viscosity representing under-resolved eddies, approximated as

$$\nu_{\text{turb}} = \Delta \cdot UL, \quad (80)$$

where Δ is a parameter related to the resolution of the mesh. In the limit of infinite Reynolds number, where $\nu_{\text{mol}} = 0$, we have $1/Re = \Delta$ and the momentum equation (77) becomes

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + \Delta \cdot \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (81)$$

5.3 Fourier representation

Using the Fourier convention (19), the viscous term becomes

$$\widehat{\frac{\partial^2 u_i}{\partial x_j \partial x_j}} = -k^2 \hat{u}_i, \quad (82)$$

and the pressure gradient becomes

$$\widehat{\frac{\partial p}{\partial x_i}} = ik_i \hat{p}. \quad (83)$$

The convective term

$$\hat{N}_i = u_j \widehat{\frac{\partial u_i}{\partial x_j}}, \quad (84)$$

will be evaluated pseudo-spectrally. The scaling terms transform as in (22) with $d = 3$, so the transform of (81) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + 3 \hat{u}_i \right) - \sigma_U \hat{u}_i - \hat{N}_i - ik_i \hat{p} - \Delta \cdot k^2 \hat{u}_i, \quad (85)$$

and the transform of the continuity equation (78) is

$$k_i \hat{u}_i = 0. \quad (86)$$

5.4 Pressure and the solenoidal projection

Contracting (85) with k_i and using the continuity constraint gives

$$0 = -\sigma_L \left(k_i k_j \frac{\partial \hat{u}_i}{\partial k_j} + d k_i \hat{u}_i \right) - \sigma_U k_i \hat{u}_i - k_i \hat{N}_i - ik_i^2 \hat{p} - \Delta \cdot k^2 k_i \hat{u}_i. \quad (87)$$

The remaining scaling contribution also vanishes: differentiating (86) with respect to k_j gives

$$\frac{\partial}{\partial k_j} (k_i \hat{u}_i) = \hat{u}_j + k_i \frac{\partial \hat{u}_i}{\partial k_j} = 0 \implies k_i \frac{\partial \hat{u}_i}{\partial k_j} = -\hat{u}_j, \quad (88)$$

so that $k_i k_j \partial \hat{u}_i / \partial k_j = -k_j \hat{u}_j = 0$. Hence the scaling terms are divergence free on their own — the self-similar frame does not generate pressure — and (87) reduces to a Poisson equation for the pressure,

$$\mathrm{i} k^2 \hat{p} = -k_i \hat{N}_i \implies \hat{p} = \mathrm{i} \frac{k_l \hat{N}_l}{k^2}, \quad k \neq 0, \quad (89)$$

with $\hat{p}$ at $k = 0$ fixed arbitrarily, since only its gradient enters the momentum equation. Substituting (89) into (85), the pressure term becomes

$$-\mathrm{i} k_i \hat{p} = \frac{k_i k_l}{k^2} \hat{N}_l, \quad (90)$$

which combines with $-\hat{N}_i$ to give the solenoidal projection of the convective term. Defining the projection tensor

$$P_{il}(\mathbf{k}) = \delta_{il} - \frac{k_i k_l}{k^2}, \quad (91)$$

the evolution equation splits exactly along the lines of the lemma (7),

$$\frac{\partial \hat{u}_i}{\partial t} = \underbrace{-\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + 3 \hat{u}_i \right) - \sigma_U \hat{u}_i}_{\text{scaling terms}} - \underbrace{P_{il}(\mathbf{k}) \hat{N}_l - \Delta \cdot k^2 \hat{u}_i}_{\text{physical terms}}. \quad (92)$$

5.5 The physics step

The physics operator (9) is the flow of the bracketed physical terms of (92),

$$\Phi_{\Delta t}^{(P)}(\hat{u}_i(t)) = \hat{u}_i(t) - \int_t^{t+\Delta t} \left[P_{il}(\mathbf{k}) \hat{N}_l + \Delta \cdot k^2 \hat{u}_i \right] dt. \quad (93)$$

Open. Unlike the 1D case, (93) has no closed form: the convective term is nonlinear, so $\Phi^{(P)}$ must be realized by an actual time-marching scheme. That scheme has not been chosen. Whatever is chosen must meet the accuracy requirement of §2.4 — second order for the Strang composition to retain its order — and must handle the dealiasing of the pseudo-spectral product (84). The linear term $-\Delta k^2 \hat{u}_i$ admits an exact integrating factor and need not be marched with the rest.

5.6 Constraint functionals

The moments (60) used for the 1D equation do not carry over: a velocity field in a periodic box has $\int u_i d\mathbf{x} \approx 0$, so F_1 as written is not usable as a scale. We therefore take the corresponding moments of the energy density $u_i u_i$, which is non-negative and so cannot cancel,

$$F_1[u] = \int u_i u_i d\mathbf{x} \equiv E_0, \quad F_2[u] = \int r^2 u_i u_i d\mathbf{x} \equiv E_2, \quad (94)$$

where $r^2 = x_j x_j$ is the squared distance from the centre of the domain (33). Thus E_0 is twice the total kinetic energy and E_2/E_0 the energy-weighted mean square radius, so E_0 measures the amplitude of the field and E_2 its spread, exactly as M_0 and M_2 do in the 1D case.

Writing E_0^*, E_2^* for the values carried by the field being scaled and substituting $\mathbf{y} = \mathbf{x} \tilde{L}$ in each integral, with $d = 3$,

$$\int \frac{1}{\tilde{U}^2} u_i u_i(\mathbf{x} \tilde{L}) d\mathbf{x} = \frac{1}{\tilde{U}^2 \tilde{L}^3} E_0^*, \quad (95)$$

$$\int r^2 \frac{1}{\tilde{U}^2} u_i u_i(\mathbf{x} \tilde{L}) d\mathbf{x} = \frac{1}{\tilde{U}^2 \tilde{L}^5} E_2^*, \quad (96)$$

so the homogeneity exponents of (24) are

$$(p_1, q_1) = (2, 3), \quad (p_2, q_2) = (2, 5), \quad (97)$$

with $p_1 q_2 - p_2 q_1 = 4 \neq 0$, so the constraint system (25) is uniquely solvable. Solving it,

$$\tilde{L} = \sqrt{\frac{E_0 E_2^*}{E_0^* E_2}}, \quad (98)$$

$$\tilde{U} = \left(\frac{E_0^*}{E_0}\right)^{1/2} \tilde{L}^{-3/2} = \left(\frac{E_0^*}{E_0}\right)^{5/4} \left(\frac{E_2}{E_2^*}\right)^{3/4}. \quad (99)$$

The length ratio (98) is identical in form to its 1D counterpart (64): in both cases $\tilde{L}$ is the square root of a ratio of mean square widths.

Open. Two properties of this pair are assumptions rather than consequences. First, the homogeneity (24) is exact only for integrals over $\mathbb{R}^d$; on the finite box the substitution $\mathbf{y} = \mathbf{x} \tilde{L}$ rescales the domain of integration as well, so (95) and (96) hold only to the extent that the energy is concentrated well inside the box. This is the same assumption the 1D moments make, and like it is not currently monitored. Second, r is measured from a fixed origin and is not periodic, so E_2 is meaningful only for a field that stays centred and localized: for a statistically homogeneous field filling the box, $u_i u_i$ is uniform on average and E_2/E_0 reduces to the mean square radius of the box itself, carrying no information about the field. Which of these two regimes the intended runs occupy should be settled before the pair is used.

5.7 Boundary treatment

The domain is a periodic box, so no boundary condition is imposed beyond periodicity itself and the substep APPLYBC of §2.4 is the identity. The tension noted in §4.5 between a spectral discretization and a clamped edge therefore does not arise here.